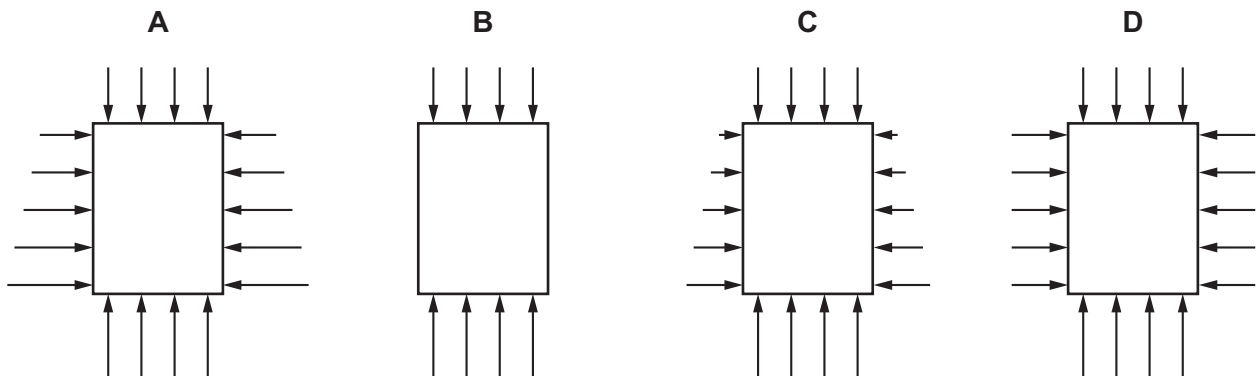
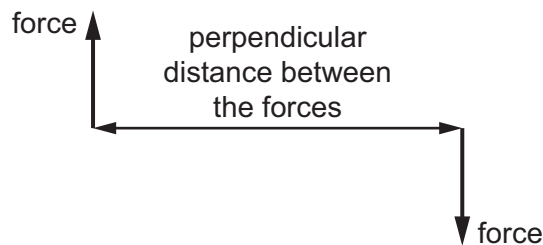


- 16** A block is submerged vertically in a liquid. The four diagrams show the block viewed from the side.

Which diagram shows, to scale, the forces exerted on equal areas of the block by the liquid?



- 17** The diagram shows a couple.



How is the torque of the couple calculated?

- A** $\frac{1}{2} \times$ perpendicular distance between the forces $\times$ magnitude of one of the forces
- B** perpendicular distance between the forces $\times$ magnitude of one of the forces
- C** perpendicular distance between the forces $\times$ magnitude of the sum of the forces
- D** $2 \times$ perpendicular distance between the forces $\times$ magnitude of one of the forces
- 18** A ball of mass 1.2 kg travels horizontally at a speed of 3.0 m s^{-1} .

The ball hits a cushion and comes to rest over a horizontal distance of 0.020 m .

What is the work done by the cushion on the ball to bring the ball to rest?

- A** 0.24 J **B** 1.8 J **C** 5.4 J **D** 11 J